Source and Diffuse-LyC Photon-Energy Sampling: Code Comparison

1. Scope

This report is based on direct source inspection of MOCASSIN 2.02.73.2, `Wood_Codes/ionize`, CMacIonize-master, `TORUS/torus/src`, and MoCHII. For equal-energy packets the source photon CDF is normally proportional to $L_\nu/(h\nu)$. A free-bound LyC CDF also follows the Milne relation and includes the inverse photoionization cross section.

2. Comparison

Code	Source energy	Diffuse LyC energy	Result
MoCHII	Continuous input-spectrum inverse CDF; sampled energy retained in continuous mode.	Temperature-indexed Milne CDF per ground continuum, inverted once after interpolating in $\log T$; separate He I cascade branches.	Exact transport energy; exact ground-continuum shape. He I cascade exact in branch weights, two-photon shape and 584 Å escape-probability branching, whose products are then transported (§7.).
MOCASSIN	Inverse CDF on global frequency grid.	Cell-emissivity CDF on the same grid.	Flexible tabulated emissivity, but grid-valued packets.
Wood ionize	Log-interpolated Planck/atmosphere CDF.	Temperature-indexed, cross-section-weighted Milne CDF.	Explicit H/He free-bound continuum shape.
CMacIonize	Numerical spectrum-specific CDF.	Dedicated H and He LyC spectra selected by reemission branches.	Modern event-by-event physical reemission.
TORUS	Tabulated wavelength CDF, linear in selected interval.	No active HII diffuse-LyC reemission path found.	Generic continuum sampler only.

3. MoCHII

`src/energy_sampler_mod.f90` constructs the source CDF and `src/main.f90` launches packets. In `ion_energy_mode = 'continuous'`, `assign_ion_packet_energy` preserves the sampled energy; `ion_packet_mod.f90` and `ion_score_mod.f90` evaluate H, He, and metal physics at that energy. The ionizing-band index is diagnostic only. `src/diffuse_mod.f90` builds a leaf–channel CDF each iteration. Its H II→H I(1s), He II→He I(1^1S), and He III→He II(1s) channels use $\alpha_1 = \alpha_A - \alpha_B$ and, since 2026-07-29, sample the Milne photon spectrum $p(E) \propto E^2 \sigma_{\rm pi}(E) e^{-(E-E_{th})/kT_e}$ through `src/milne_recomb_spectrum_mod.f90`, selected by `par%diffuse_energy_model = 'milne'` (the default). The tables are built once in $x = (E - E_{th})/kT$, interpolated in $\log T$, and inverted once; both pseudorandom and Sobol launches call the same routine, and the Sobol dimension assignment is unchanged. Because packets are equal-energy and their count comes from the channel's energy luminosity, they are drawn from $E\,n(E)$ rather than $n(E)$, which makes the represented photon rate exactly $n_e n_{\rm ion}\alpha_1 V$; the luminosity still uses the number-weighted mean $\langle E\rangle_n$. Start-up reports α_1 recovered from the same spectral integral against $\alpha_A - \alpha_B$ without rescaling either. `exponential` ($E = E_{th} + kT_e[-\ln u]$) is retained for reproducing earlier runs and `threshold` as a sensitivity bracket; neither is a physical replacement. The optional He I cascade now weights its 19.82-eV, 21.22-eV and two-photon branches by effective recombination coefficients instead of the AGN3 statistical weights, and draws its ionizing two-photon energies from the Drake et al. (1969) shape rather than uniformly. At 8000 K the branch fractions are 0.762 : 0.076 : 0.162 against the statistical 0.750 : 0.083 : 0.167, a temperature-dependent correction rather than a reordering, and the channel radiates 19.223 eV in 0.9663 photons per case-B He II recombination instead of the 19.200 eV in 0.9634 photons of the statistical weights with a flat two-photon spectrum (+0.12% in energy, +0.30% in photon number), before the 584 Å branching of §7. is applied on top. Section 7. compares this cascade, and the fate of its 584 Å photon, with the three reference codes.

4. MOCASSIN

In `source/photon_mod.f90`, `newPhotonPacket` calls `getNu2(inSpectrumProbDen,...)` for stellar photons. `getNu` and `getNu2` invert cumulative arrays and return an integer frequency pointer. `continuum_mod.f90` builds the continuum representation and `setLdiffuse` accumulates diffuse luminosity. Diffuse continuum packets use the same pointer/CDF mechanism. Thus arbitrary free-bound emissivities can be tabulated, but packet frequency is a global-grid value rather than a continuous in-bin variate.

5. Wood ionize

`planckset.f` prepares the source CDF; `nuset.f` locates a uniform deviate in `bbcdf` and logarithmically interpolates `lognu`. `probinit.f` explicitly constructs photon CDFs, $\int j_\nu/\nu\, d\nu$. At every `Tarray` point it calls `phfit2` and uses $j_{HI} \propto \nu^3\sigma_{HI}(\nu)\exp[-157919.667(\nu-1)/T]$, with an analogous He I expression above 1.81 Ryd. It integrates j_ν/ν to `pH` and `pHe`. `nuemit.f` inverts the temperature-indexed H/He CDFs and the He two-photon CDF; `mcionize.f` selects the reemission branch using `probset.f`. This is a Milne-grid sampler, though its output is a grid frequency.

6. CMacIonize

`PlanckPhotonSourceSpectrum.cpp` tabulates $\nu^2/(e^{h\nu/kT}-1)$ over 1–4 Ryd, integrates a CDF, and interpolates in logarithmic CDF–frequency space in `get_random_frequency`. `PhysicalDiffuseReemissionHandler.cpp` chooses absorption with $n_i\sigma_i$ weights and applies temperature-dependent reemission probabilities. It invokes `HydrogenLymanContinuumSpectrum` and `HeliumLymanContinuumSpectrum`, and separately treats the 19.8-eV and 56% ionizing He two-photon branches. These dedicated spectra receive cross-section objects, implementing the same physical design as Wood ionize.

7. He I cascade and the 584 Å photon

The reemission branch that `probset.f` sets up above is where the four codes differ most, and it holds two separate decisions: which $n = 2$ term a case-B He II recombination passes through, and what becomes of the 2^1P photon, He I Lyα at 584 Å = 21.22 eV.

Branch coefficients. `probset.f` sets four effective recombination coefficients from Benjamin, Skillman & Smits (1999) Table 1 with J. S. Mathis — $1.54 \times 10^{-13} T_4^{-0.486}$ to 1^1S, $2.10 \times 10^{-13} T_4^{-0.381}$ to 2^3S, $2.06 \times 10^{-14} T_4^{-0.451}$ to 2^1S and $4.17 \times 10^{-14} T_4^{-0.695}$ to 2^1P — and normalizes by their sum, for the reason the source states: "different T-dependences in fitting formulae". `PhysicalDiffuseReemissionHandler` in CMacIonize is a line-for-line reimplementation of that choice, with the same four coefficients, the same sum normalization and the same cumulative channel order, and cites Wood, Mathis & Ercolano (2004) §3.3. MOCASSIN does not resolve the split at all: its nebular continuum uses $6.23 \times 10^{-14} T_4^{-0.827}$ for the He I two-photon emissivity under the stated assumption that "all HeI singlets finally end up in the 2^1S". That coefficient is the singlet total, $2.06 \times 10^{-14} + 4.17 \times 10^{-14} = 6.23 \times 10^{-14}$ exactly, refitted as a single power law, so it is the full-conversion limit written as a constant.

The 584 Å photon. `mcionize.f` does not propagate it. It draws $p_{\mathrm{Hots}} = [1 + 0.77(100/\sqrt{T})x(\mathrm{He}^0)/x(\mathrm{H}^0)]^{-1}$ and with that probability absorbs the photon on the spot into the hydrogen Lyman continuum, depositing $21.22 - 13.60 = 7.62$ eV; otherwise the 2^1P is collisionally converted to 2^1S and decays as the two-photon continuum, from which an ionizing photon emerges with probability 0.56. The same p_{Hots} enters the ionization balance (`h0find.f`) and the heating (`ioneng.f`). CMacIonize writes $p_{\mathrm{Hots}} = \sqrt{T}x(\mathrm{H}^0)/[\sqrt{T}x(\mathrm{H}^0) + 77\,x(\mathrm{He}^0)]$, which is algebraically the same expression, and also destroys the photon at the emission site.

MoCHII. Since 2026-07-30 the branch fractions come from the same three $n = 2$ coefficients under the same sum normalization, so all three are drawn from one source: a ratio of coefficients with unrelated temperature dependences is not a branch fraction, and closing one branch against an external α_B would place the mismatch between two data sets in whichever branch is solved last. The sum is $0.90\,\alpha_B(\mathrm{He\,II})$ at 8000 K and 0.97 at 10^4 K, and that shortfall is reported at start-up instead of being forced into a branch. The two-photon energies follow the Drake table (0.55644 ionizing photons and 8.9646 eV per decay, in-band mean 16.110 eV) rather than the flat surrogate (0.56, 17.109 eV). The 2^1P branch is split by the same escape probability as in the two reference codes, written in this module as $p_{584} = [1 + 77\,x(\mathrm{He}^0)/(\sqrt{T}\,x(\mathrm{H}^0))]^{-1}$; the gate `tests/hei_cascade/check_hei_584.f90` shows this form agreeing with Wood's and CMacIonize's algebraically distinct ones to 2.2×10^{-16}, both limits exact, and $p_{584} = 1$ reproducing the previous branch energies. Because p_{584} depends only on the emitting leaf's own state it enters the branch weights of the emission table, so no Sobol dimension was added and the assignment is unchanged. A converted decay radiates 20.62 eV as the two-photon pair rather than 21.22 eV, the missing 0.60 eV leaving as the non-ionizing 2.06 micron $2^1\mathrm{P} \rightarrow 2^1\mathrm{S}$ line. The same split now sets the He I two-photon coefficient of `nebcont_mod`, $\alpha_{2q} = \alpha(2^1\mathrm{S}) + [1 - p_{584}]\alpha(2^1\mathrm{P})$, which reduces to MOCASSIN's 6.23×10^{-14} in the $p_{584} \rightarrow 0$ limit; the transported packets and the written continuum therefore describe the same nebula, closing an internal disagreement in which the two modules had sat at opposite limits of the same decay. (An earlier version of this report read 6.23×10^{-14} as the 2^1S coefficient alone and closed 2^1P against α_B, obtaining $0.746 : 0.224 : 0.031$ and an argument that case-B trapping inverts the singlet split. That reading is withdrawn: the coefficient is the singlet total.)

Code	$n = 2$ branch coefficients	2^1P (584 Å) fate	Transported?
Wood ionize	BSS99 + JSM, four coefficients, normalized by their sum	p_{Hots} ionizes H on the spot, $1 - p_{\mathrm{Hots}}$ converts to 2^1S two-photon (ionizing fraction 0.56)	No (on the spot)
CMacIonize	Identical set and sum normalization, citing WME04 §3.3	Identical branching, written in an algebraically equivalent form	No (on the spot)
MOCASSIN	Singlet total $6.23 \times 10^{-14} T_4^{-0.827}$ only	All singlets assumed converted to 2^1S	Emissivity only
MoCHII	Same three $n = 2$ coefficients, normalized by their sum; $\sum \alpha/\alpha_B$ reported, not forced	Same escape-probability split, from the leaf's own state; converted decays radiate 20.62 eV with the Drake two-photon shape, and `nebcont_mod` uses the same split	Yes (AMR grid)

Where MoCHII stands. Of the four codes only MoCHII both applies the branching and then transports what comes out of it: Wood and CMacIonize destroy the 584 Å photon where it is emitted, so its deposition is local by construction, and MOCASSIN carries the decay in an emissivity only. MoCHII draws the same split from the same closure and hands the surviving 584 Å packets and the two-photon photons to the grid, so where their energy is deposited is resolved rather than assumed. The branch content is no longer the weak part of that arrangement: the branch fractions, the two-photon shape and the 584 Å fate all come from the same sources the reference codes use. Evaluated on the converged production grids and weighted by the 2^1P emission rate, p_{Hots} is 0.117 at HII40 and 0.0044 at HII20, so 88% and 99.6% of the branch converts to the two-photon continuum, and the channel loses 7–8% of its photons and softens. What remains here is explicit 584 Å resonance-line transfer, for which p_{Hots} is an escape-probability closure valid for an optically thick, static, dust-free line, and collisional redistribution among the $n = 2$ terms at high density.

8. TORUS

`spectrum_mod.f90:getWavelength` draws r, locates it in `spectrum%prob`, and linearly interpolates adjacent tabulated wavelengths; `photon_mod.f90:initPhoton` uses it. Searches of active `torus/src` found no Wood/CMacIonize-like H–He reemission handler, temperature-indexed free-bound LyC CDF, or HII recombination-ionizing-packet emission call. This is an absence of exposed HII diffuse-LyC physics, not evidence for a threshold approximation.

9. Implication

MoCHII already evaluates transport cross sections at each sampled source energy, unlike frequency-pointer approaches. Its ground-state diffuse continuum, formerly the distinct limitation, now follows the Milne relation with the same threshold-aware cross section transport uses, interpolating the CDF in temperature before a single inversion — the prescription §10. argues for, and one step beyond both Wood (nearest temperature row, grid frequency) and CMacIonize (interpolation between frequencies drawn from two temperature rows). The measured outcome confirms the prediction made here: the converged HII40 calculation moves the He I front by only +0.00063 pc, closing 0.3% of the Cloudy residual. The exact sampler was physically required and it did not remove the residual. Three further corrections have followed inside the He I excited cascade: the Drake two-photon shape, −0.00030 pc; one consistent set of branch coefficients, +0.00024 pc; and the 584 Å escape-probability split, −0.00363 pc, the largest of the four and six times the Milne step. Four independent improvements to what the diffuse field *emits* are therefore worth −0.00306 pc in total, while the channel's mere presence is worth 0.10 pc: $r(\mathrm{He}^0 = 0.5)$ moves from 4.12333 to 4.11970 pc against the Cloudy c25.00 value of 4.30531 pc, so the sum is 1.7% of the 0.18 pc gap and points away from Cloudy rather than toward it. Each of the four is separately required — the acceptance criterion is physical correctness, not agreement with a particular reference calculation — and together they settle the diagnosis: **the residual is not in what the diffuse field emits.** The shape of the emitted spectrum, its branch weights and the fate of its strongest line have each been corrected independently, against an external implementation or an exact relation, and their sum is negligible and of the wrong sign. What remains, in order of plausibility, is the *transport* of those photons, the one axis never varied in this investigation; the He I ground-continuum atomic data, where α_1 from the Milne integral and the fitted $\alpha_A - \alpha_B$ disagree by 2–3.4% for He I against 0.5% for H I and 0.4% for He II, in exactly the continuum whose front is discrepant; collisional redistribution among the $n = 2$ terms; and a difference outside the diffuse field altogether.

10. Which method is physically most accurate?

The most accurate practical Monte-Carlo prescription is *not* a fixed energy, an exponential tail, or merely a very fine arbitrary energy grid. For each recombining ion, electron temperature, and (where required) density, construct the photon-number PDF from the level-resolved free–bound emissivity obtained by detailed balance,

$$p_i(\nu \mid T_e) \propto \frac{j^{\mathrm{fb}}_{\nu,i}(T_e)}{h\nu}, \qquad \nu \geq \nu_{0,i},$$

using the same resolved photoionization cross section in the Milne relation that is used to absorb the packet. The emissivity should include the relevant free–bound Gaunt factor and should be integrated with accurate quadrature; its CDF and its mean photon energy must be derived from that same PDF. Branching to recombination levels and subsequent He I/He II cascades must be selected from level-resolved effective coefficients, with the full two-photon spectral shape rather than a flat surrogate. This prescription is exact subject to the adopted atomic data and the assumption of a Maxwellian electron distribution. Wood ionize and CMacIonize are closest among the reviewed codes: both use $\nu^3\sigma(\nu)\exp[-h(\nu-\nu_0)/kT]$ and temperature tables. CMacIonize improves the source CDF interpolation, but its LyC implementation interpolates frequencies selected from two neighboring temperature CDFs rather than directly in a temperature-interpolated CDF; Wood uses the nearest temperature row. Therefore neither is the final reference implementation. MoCHII now does interpolate the *CDF* in temperature, invert once, sample a continuous energy, and carry that same energy into the existing exact transport cross sections. Its He I cascade now uses one consistent set of effective recombination coefficients, the measured two-photon shape and the same 584 Å escape-probability split as Wood and CMacIonize, applied before transport rather than in place of it (§7.), so the remaining approximations there are explicit resonance-line transfer of the trapped 584 Å photon and collisional redistribution among the $n = 2$ terms, not the shape of the emitted continua.

11. Why can Wood ionize be fast?

The accuracy of the diffuse-LyC *spectral shape* is not the dominant cost. Wood ionize precomputes its small T–ν Milne CDF arrays in `probinit.f`; during transport, `nuemit.f` only draws a uniform deviate and performs a table search. It does not numerically integrate the Milne relation or evaluate a new continuum spectrum for every emitted packet.

More importantly, Wood uses event-by-event reemission. Following absorption, `mcionize.f` selects absorption species and reemission channel; a diffuse packet is created only when that event branches into an ionizing reemission. MoCHII instead rebuilds an explicit volume-emissivity CDF every iteration and transports a separate diffuse population whose size is $L_{\mathrm{diffuse}}/L_{\mathrm{packet}}$. In HII40 this is commonly 10^7–2×10^7 diffuse packets per iteration, in addition to direct packets. Wood also confines this benchmark physics largely to H and He and samples a modest frequency grid. MoCHII additionally pays for AMR leaf traversal, continuous-energy H/He/metal path scoring, metal ionization cascades and cooling, thermal equilibrium iterations, and HDF5 diagnostics. These choices can be scientifically necessary, but they dominate runtime relative to one inverse-CDF operation.

Consequently, adding a precomputed Milne CDF to MoCHII should have negligible direct sampling cost. It should not be expected by itself to make MoCHII as fast as Wood: the major cost is explicit diffuse-packet transport and the broader MoCHII physics. The correct comparison is therefore limited: Wood is closest for the H/He ground-continuum spectral PDF, but its nearest-temperature, grid-frequency sampling is still less rigorous than a MoCHII CDF/PDF-temperature interpolation followed by continuous-energy inversion.